Your grade: 93.33%

Your latest: **93.33%** • Your highest: **93.33%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What does a high constraint value indicate about a node? 1 / 1 point

- The node is isolated from all clusters
- The node's connections are concentrated within a single group
- The node connects to many distinct communities
- The node has no ego-network

Nice work

Feedback: High constraint means the node is embedded in one cluster and not acting as a broker.

2. What is the **0-core** of a network? 0 / 1 point

- Only nodes with at least one neighbor
- A set of isolated nodes
- Nodes with no connections
- The entire set of nodes in the network

Try again

Feedback: This applies to the 1-core, not the 0-core.

3. What would a negative assortativity coefficient indicate? 1 / 1 point

- Nodes are equally connected to all other nodes
- No correlation between node types
- Perfect assortativity in the network
- Disassortativity, where nodes tend to connect to dissimilar nodes

Nice work

Feedback: A negative coefficient suggests a preference for dissimilar node connections.

4. Which node type is most likely to bridge a structural hole? 1 / 1 point

- A node with high constraint
- A broker
- A peripheral node
- A leaf node

Nice work

Feedback: Brokers connect otherwise disconnected clusters, thus bridging structural holes.

5. What does the "effective size" of a node measure? 1 / 1 point

- The closeness of a node to the network center
- The number of nodes in the entire network
- The node's degree adjusted for redundancy in its ego-network
- The influence of a node based on content sharing

Nice work

Feedback: Effective size accounts for both the quantity and uniqueness of a node's connections.

6. Which of the following statements about ego-networks is correct? 1 / 1 point

- The ego-network includes only the ego and the nodes it connects to
- Ego-networks are used to calculate assortativity
- An ego-network includes all nodes and edges in the full network
- An ego-network includes the ego's neighbors and the connections between them

Nice work

Feedback: Correct – an ego-network contains both neighbors and their interconnections.

7. What is the **k-core** of a network? 1 / 1 point

- A group of nodes with the same degree
- A maximal group of nodes where each node has at least k connections within the group
- A group of nodes that form the shortest path tree
- The subset of nodes with the highest centrality

Nice work

Feedback: Correct. The k-core includes all nodes with at least k neighbors in the same group.

8. In the context of assortativity in undirected networks, which of the following are valid approaches? 1 / 1 point

- [x] Converting undirected edges into two directed edges going in opposite direction

Nice work

Feedback: This allows you to use the same measure on undirected graphs.

- [ ] Using undirected graphs without modification
- [ ] We cannot measure assortativity in undirected networks

9. Which of the following best explains the mechanism of selection in homophily? 1 / 1 point

- People are assigned randomly to network clusters
- People form connections with those who are already similar
- People choose to avoid others with differing interests
- Friends influence each other to adopt similar habits

Nice work

Feedback: Selection describes choosing connections based on pre-existing similarities.

10. Which of the following are **properties** of a k-core? 1 / 1 point

- [ ] K-cores must always be connected components
- [ ] Nodes with degree less than k can still be in the k-core
- [x] K-cores are nested subgraphs

Nice work

Feedback: Correct. The (k+1)-core is always a subset of the k-core.

- [x] A node in a k-core must have at least k neighbors within the same core

Nice work

Feedback: Correct. That's the core definition of a k-core.

11. Which statistical measure is used to calculate numerical assortativity in a network? 1 / 1 point

- Pearson correlation between source and target node attributes
- Mean degree of the node
- Median of node values
- Standard deviation of node attributes

Nice work

Feedback: This statistical method helps quantify the similarity between connected nodes' numerical attributes.

12. Which of the following statements is true about **k-core decomposition**? 1 / 1 point

- It identifies the shortest path between all core nodes
- It calculates influence spread across a network
- It splits a network into disconnected components
- It identifies nested layers of cores from peripheral to central

Nice work

Feedback: Correct. K-core decomposition reveals core-periphery structure by progressively peeling away lower-degree nodes.

13. What does the term "homophily" describe in the context of social networks? 1 / 1 point

- The tendency of nodes to avoid similar connections
- The division of nodes into completely disconnected clusters
- The tendency of nodes to connect with similar others
- The tendency of nodes to connect randomly

Nice work

Feedback: Homophily refers to the preference of nodes to connect with other nodes that have similar attributes.

14. Which of the following statements are true about brokers in a network? 1 / 1 point

- [ ] Brokers are nodes with the highest number of total connections
- [x] Brokers help close structural holes between network clusters

Nice work

Feedback: This is one of the defining roles of a broker.

- [x] Brokers can access diverse information from multiple groups

Nice work

Feedback: Brokers benefit from exposure to varied information sources.

- [ ] Brokers are always located at the periphery of the network

15. Why is **k-core decomposition** useful in network analysis? 1 / 1 point

- It eliminates redundant edges from the network
- It provides the shortest paths between all core nodes
- It assigns weights to edges for better clustering
- It identifies structurally cohesive groups of nodes

Nice work

Feedback: Correct. K-core helps detect cohesive layers within a network.